Siqiao Huang

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Education

- **IIIS (Yao Class), Tsinghua University** *2023 - 2027 (expected)*
  B.S. in Computer Science and Technology; Advisor: Prof. Hang Zhao, Prof. Mingsheng Long. Beijing, China
  - GPA: 3.94/4.00; Rank: 9/91
- **Machine Learning Department, Carnegie Mellon University** *2025 Jul. - Sep.*
  Visiting Research Intern; Advisor: Prof. Max Simchowitz. Pittsburgh, USA
- **Paul G. Allen School, University of Washington** *2026 Apr. - Sep.*
  Visiting Research Intern; Advisor: Prof. Abhishek Gupta. Seattle, USA

Research Interests

- My research aims to **develop models with intrinsic understandings of the physical world** and apply them to enable **general-purpose decision-making**. My research focuses on:
  - **World Models for Decision-Making:** *I aim to build the next generation of predictive dynamics models, and applying them for decision-making through model-based reinforcement learning, post-training policy models in world models, and test-time planning.* Related past work: Vid2World, NanoWM, and TrajWorld.
  - **Pretraining of Generalist Embodied Policies:** *I hope to understand how we can build generalist policies that understand the world through pretraining, with a focus on data-centric and representation-centric approaches that leverage visual and dynamical priors.* Related past work: OpenWAM, OMG, and WB-WAM.
  - **Adaptation and Self-Improvement of Robot Policies:** *I aspire to bake world understanding into policies through physical interaction, including DAgger, real-world RL, and simulation post-training. Besides learning from visual worlds (real, simulated, or learned), I am very intrigued by leveraging language-centric models for self-improvement in the physical world.* Related past work: MLS-Bench, HarnessEval-W, and SURGE.

Publications

* equal contributions, † corresponding author

[1] **OMG: Omni-Modal Motion Generation for Generalist Humanoid Control**. [🌐] [📄] [GitHub]
Siqiao Huang*, Kun-Ying Lee*, Dongming Qiao*, Guanqi He*, . . . , Hang Zhao†.
arXiv preprint, 2026; ExWBC @ RSS 2026 (Oral).

[2] **Nano World Models: A Minimalist Implementation of Future Video Prediction**. [🌐] [📄] [GitHub]
Siqiao Huang, Partha Kaushik, Michael Chen, Hengkai Pan, . . . , Max Simchowitz†.
Technical report, 2026.

[3] **Vid2World: Crafting Video Diffusion Models to Interactive World Models**. [🌐] [📄] [GitHub]
Siqiao Huang*, Jialong Wu*, Qixing Zhou, Shangchen Miao, Mingsheng Long†.
ICLR 2026.

[4] **OpenWAM: An Open, Modular Exploration Towards Systematic World–Action Model Pretraining**. [🌐] [📄] [GitHub]
Yuran Wang*, **Siqiao Huang***, Mingleyang Li*, Chenhao Zhang*, Jiaqi Liang*, . . . , Xiaowei Chi, Guanqi He, Shanghang Zhang, Hao Dong, Lin Shao†, Hang Zhao†.
arXiv preprint, 2026.

[5] **MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI**. [🌐] [📄] [GitHub]
Bohan Lyu*, Yucheng Yang*, **Siqiao Huang***, . . . , Jiantao Jiao, Dawn Song, Chi Jin†.
NeurIPS 2026, Evaluations & Datasets Track (Oral, 15/3,757).

[6] **SURGE: On the Potential of Large Language Models as General-Purpose Surrogate Code Executors**. [📄] [GitHub]
Bohan Lyu*, **Siqiao Huang***, Zichen Liang*, Qian Sun, Jiaming Zhang.
***EMNLP 2025, Main Conference**; top 0.3% meta score.*

[7] **GroundingPI: A Grounding Foundation Model towards Physical Intelligence with Visual Primitives**. [🌐] [📄] [GitHub]
Qize Yu*, Lianrui Fan*, . . . , **Siqiao Huang**, . . . , Ping Luo†, Shiyu Huang†.
arXiv preprint, 2026.

[8] **WB-WAM: Heterogeneous Body–Hand Pre-training for Humanoid Loco-Manipulation**. [🌐] [📄]
Chuan Qin*, Shaoting Zhu*, Siyuan Luo, **Siqiao Huang**, Hongyu Zhao, Hang Zhao†.
arXiv preprint, 2026.

[9] **HarnessEval-W: Agentifying the Evaluation of Visual Worlds**. [🌐] [📄] [💻]
MirroS Team: Weiliang Chen*, Haowen Sun*, Jun Gao*, . . . , **Siqiao Huang**, . . . , Fangfu Liu.
Technical Report, 2026.

[10] **Trajectory World Models for Heterogeneous Environments**. [📄] [💻]
Shaofeng Yin*, Jialong Wu*, **Siqiao Huang**, Xingjian Su, Xu He, Jianye Hao, Mingsheng Long†.
ICML 2025.

Selected Projects

- **Awesome World Models** *Oct 2025- Dec 2025*
  3.4k GitHub Stars | Curated Research and Resources on World Models [💻] [🌐]
  ◦ A curated collection of world-model research and resources for researchers and practitioners.
  ◦ Sole creator and core maintainer of the GitHub repository, with **3.4k stars**.

Research Experience

- **WEIRD Lab, University of Washington** *Apr 2026 - Present*
  Advisor: Prof. Abhishek Gupta | Topics: Real-World RL, World Models, Test-time Adaptation
  ◦ Sample-Efficient Fleet-level Real-World Reinforcement Learning via Coordination

- **MARS Lab, Tsinghua University** *Nov 2025 - Present*
  Advisor: Prof. Hang Zhao | Topics: Dexterous Manipulation, WAM, Humanoid Foundation Models
  ◦ OMG-Posttrain: Post-training for General but Agile Humanoid Foundation Models.
  ◦ DS-GRPO: Noise-Space Adaptation of Diffusion Policies with Action-Space Verification.
  ◦ OMG: Omni-Modal Motion Generation for Generalist Humanoid Control.

- **Machine Learning Department, Carnegie Mellon University** *July 2025 - Apr 2026*
  Advisor: Prof. Max Simchowitz | Topics: World Models, Visual & Action Tokenization
  ◦ Compression Forcing: Hierarchical Tokenization Enables Flexible World Models.
  ◦ Nano World Models: A Minimalist Implementation of Future Video Prediction.
  ◦ MLS-Bench: A Holistic and Rigorous Assessment of AI Systems on Building Better AI.

- **THUML Group, Tsinghua University** *Jul 2024 - Jul 2025*
  Advisor: Prof. Mingsheng Long | Topics: World Models, Dynamical Systems
  ◦ Vid2World: Crafting Video Diffusion Models to Interactive World Models.
  ◦ Trajectory World Models for Heterogeneous Environments.

Industry Experience

- **Wuji Technology**
  ◦ Research Scientist, Post-training Group | Topics: Simulation RL, Self-Play *Sep 2026 - Present*
  ◦ WAM Team Lead, Pretraining Group | Topics: World Action Models *Jan 2026 - Apr 2026*

Selected Honors and Awards

- **SenseTime Scholarship** *Jun 2026*
  SenseTime Technology, ***Top 30 Undergraduates in Artificial Intelligence Nationwide***

- **Comprehensive Excellence Award** *Nov 2024, 2025*
  Tsinghua University, University Scholarship

Professional Services

- **Reviewer**
  Conferences: ICRA 2027, ICLR 2027, NeurIPS 2026, CoRL 2026.

- **Teaching Assistant for "Introduction to Artificial Intelligence"**
  Introduction to Artificial Intelligence, Spring 2025, Tsinghua University. Instructor: Prof. Mingsheng Long.
  ◦ ***First and Only Undergraduate TA.*** *Graduate-voted Favorite Course Award* ***(Top 1%, 20/4000+)***.

Skills

- **Language:** TOEFL: **6.0/6.0 (On First Trial)**, CET-4: 688/710, CET-6: 685/710.